Active InferAnt Stream 004.1 ~ Does Synthetic Fabric dream of Active Inference (too)?

> **Source**: YouTube **Playlist**: Active InferAnt Stream **Duration**: 1:29:56 | **Uploaded**: Unknown **Transcript method**: auto_caption

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all right hello and welcome everyone this is active infer an stream number 4.1 it's October 2nd 2024 and as we often do we'll start it out with a GitHub push oh it went to three why not go with it this is getting pushed to the active inference Institute GitHub to the repo named fabric this is a fork of Daniel miser's collaborative project called fabric so let's verify that we got everything updated let's just double check that we're in and then we will okay that went to fabric just want to make sure that we got it going on the code here okay H interestingly not there well we'll we we we'll check in we'll check in later um this will be a short interesting stream of course open to anyone else's suggestions going to look at this fabric framework work for orchestrating llm and other kinds of synthetic interoperability and just look at a few of the functionalities that I've been playing around with today so it'll be rough it's 5:00 p.m. but so it goes okay first I'm using cursor down the bottom of the page you can see the change log I'm using cursor 0.413 um we're going to be looking at the fabric framework it took a little bit of playing around with goang and the installation environment to get fabric running but it's running it's a beauty it's fun to work with and we're going to be playing around with it and the output where this really gets one to is here's live stream let's say 57.2 here we have a quiz on active inference and re relevance realization we have the main idea being output we have the predictions testable predictions with confidence extraction of questions extraction of wisdom agents extract wisdom output improve academic writing the metadata for the YouTube video summarizing The outputs of the whole video the transcript of the video a tweet a hacker one report and a micro essay and those are just a few of the patterns so let's pull back talk about the context talk about fabric jump into it and then also try to find out what is really happening with this what is happening maybe it wasn't pushed there we go there's the Fran stream 4 okay it was there there was just a glitch in The Matrix okay in the output folder that's where everything goes so let let's uh jump in with seeing what it looks like when it's running and then we'll pop back to what fabric can do and then we'll play around with like a few next steps because there's so many cool things that can happen um let's look at the fabric documentation just briefly first okay so also for those who are watching live please write any comments or questions in the live chat it it'll be hilarious and awesome to to see what people throw into the mix okay fabric is an open source framework for augmenting humans using AI there's introduction videos what and why since the start of 2023 in generative AI we've seen a massive number of AI applications for accomplishing tasks it's powerful but it's not easy to integrate this functionality into our lives in other words AI does doesn't have a capabilities problem it has an integration problem fabric was created to address this by enabling everyone to granularly apply AI to Everyday challenges philosophy AI isn't a thing it's a magnifier of a thing or process and that thing is human creativity so let's keep that in mind and think about the flourishing of the creativity and the learning as we continue on they talk about having too many prompts and this kind of trade-off between having a libr library of prompts and copying and pasting and manually doing that and do you sort that into folders do you make a database with tags how how do you really get all those prompts or or do you meta prompt do you ask it to write a prompt that does this or that and then copy that or do you automate the asking of the meta prompt installation again it's going to be situational on your setup but it's a golang package so that's a new language for an infan stream and after getting the goang 1.23 installed getting fabric installed was pretty straightforward on my Linux computer running setup went through a a command line based setup process where I entered in API keys for open Ai and open router so there's multiple different llm handlers so that's pretty useful and then once you have it all set up here's how to use it this is the command line version and the version that I'll be showing is more running it in a script format so fabric uses what are called patterns that's really cool it's an interesting crossover or touch point with Atlas and all this other work on pattern languages um it's very intuitive and it's an exciting way to have these prompts stored and then call different patterns in uh posable ways so here's summarizing a pattern based upon an input and you can make custom patterns we might play with that later and there's some helper apps so cool thanks for keeping this an open source project let's talk about how we used it so um at The Institute as those watching on YouTube may know we have several hundred videos so this is the back end of the video production oh no no this is an Open document this is I'll even paste this into the YouTube live chat this is the back end of our video organizing so here's our our different YouTube video series so guest streams and live streams and several hundred different videos all connected up with the URL and all this kinds of information and so that's a lot of time to watch I mean even those of us who were there maybe didn't even watch the whole thing so it's like how are we going to make sense of this utilize it translate it make it more applicable make it more rigorous make it more accessible build on the past while respecting it all these other kinds of things that we like to do with archives so to do that I made the page video table for download so I used Koda to subset down to this simpler table that has just the YouTube links so it's around 500 and some number of video links what series it was in the date including the Futures the event name the guests and then the title of the strep so that was input data I uh then I'll just just for completeness options in Koda three dots on the top right download a CSV so I downloaded that exact file as CSV okay then here's extract from Institute videos well first here's the Institute C SV what it looks like so we can see there's exactly those information here's the guest stream in the future with you and here's some of the ones that have happened in the last few weeks and months um extract from Institute videos. py it's just what I'm more comfortable with so I wrote this script in Python it could be done in goang as well what this does is First Call in some imports then it selects from a set of fabric patterns to apply so we'll look at what those fabric patterns are before running the script there's the definition of some logging methods time parsing one little bump I encountered was the differences in YouTube url schema a method to get the video metadata a method that I'm not going to use right now but you can use it to download audio and visual by the way this will work with any set of YouTube URLs any um playlist you can just get all the URLs any channel so you can do this with public YouTube content I'm just doing it for the Institute content and you can download and it uses ffmpeg to convert the audio so let that one run overnight with archiving the audio and the video here's saving the transcript that's kind of a core entity and then here's the main it's going to open up CSV that we just looked at it's going to process each of the entries in the CSV create folder so here like we're looking at insights and then get and save the video metadata so let's look at insights 18 here's the metadata for insight wow 7500 seconds 1,47 views get the metadata if it's a future event there's a logged error message that it hasn't happened yet it'd be pretty funny though like what does it expect the transcript will be if the flags are set it downloads the audio and the visual saves a transcript and then iteratively applies each of the patterns saving them to the name of the pattern so again for insights 18 let's look at creating the quiz output so we we we can ask Darius and see what he thinks about how well these held up but being able to generate quiz output from arbitrary videos is pretty awesome so how how does that work with the fabric patterns and then let's play around a little bit see what people comment just see where it goes okay so that's all in the folder called Institute YouTube analysis patterns is the big folder to check out so here are all kinds of patterns that people have made and you can make your own and you can just put it into the folder so let's look at create story explanation this reads like kind of a classic concatenator prompt it starts out by telling the llm telling the agent what it is it has a 4312 IQ the goal the steps this one's pretty funny spend 2192 hours studying the content from thousands of different perspectives that's like if you if you if you mix up your PhD with a three-year-old you get something like that and then you have these more instructions and then it gives um that output so each of these patterns like let's look at create quiz this pattern generates questions to help a learner student review the main concepts of the learning objectives provided optional to be defined here in a context file so we we could subspecific run in bash so some of this information is is not even necessarily that's the read me for the The Prompt and then the system. prompt is the one that gets used so here's the one where it says okay now output this format and then that is exactly what we saw for insights 18 the quiz so that's how patterns are used in fabric hey Jeff higia maximalist so that's that's pretty cool you can create a new folder make it a new pattern like a little modular piece of intelligence work that you want to do or some task you want to do write an email and then oh well there's three subtasks there this is like cognitive industrialization what are the subtasks okay how could those be done what would be the measures would there have to be an overseer on that and um let's play around with what it looks like when the script is run pull back to extract from extract from YouTube videos is just for a single YouTube video or or a list but it does it does exactly the same thing I just made that one first made the single video situation first and then made it take in the CSV next after I knew that it would be doing fine with one okay so when we run Python 3 extract from Institute videos first it starts processing videos that are in the future it says uh the live event will begin in 143 days so it munches through the first ones in the CSV that are in the future those don't have transcripts on YouTube they shouldn't then we get to one that just happened Round Table 20243 saves a transcript and now we're getting this logging so let's look at these 202 24.3 so because I already ran it the outputs are already there but this is what it looks like to to also run it so we we'll leave it running and it might just overwrite where we're looking so let's go from the top okay first subject active inference and Institute updates learning objective understanding the structure and functions of the active inference Institute specific questions good questions to know very like thought-provoking useful questions to know and wonder about analyzing the developments and projects within the active inference ecosystem these are these are things that could be answered with an llm or they could be thrown to a person let's make a translate pattern see if we can implement it on the stream so these are the quizzes main idea active inference Institute is fostering collaborative projects and discussions to enhance applied active inference H main recommendation engage in collaborative projects and contribute to discussions for Effective Learning and Development in active inference okay predictions output let's look at this one in the YouTube or in the GitHub repo fabric Round Table predictions okay the ecosystem paper let let's see how these estimates line up the ecosystem paper will be versioned by November 24 hi the fourth applied active inference Symposium is scheduled for November 3 through 15 2024 interesting 13 through 15 actually but close enough high confidence new visuals for the ecosystem paper will be developed by the Symposium medium confidence research fellows applications will be reviewed in two weeks from October 1st medium grants for decentralized agents will be submitted by September 2024 now while that was medium in September in October it was no longer medium it was confirmed and more organizations Partnerships with organizations so just interesting let's look at more of a a Content based stream but this is just showing that even for 30 minute long update streams it's possible to make these kinds of videos let's look at a textbook group okay cohort 6 talking about chapter 5 part one look at it in GitHub okay cohort 5 sorry cohort 6 chapter 5 part one okay and if if people watching have any suggestions or questions just just go for it I just want to show a few of these moves and and meanwhile it it got through the round table it goes through all the patterns and then it goes on to the next video so I'm going to stop it the total costs though here's my open AI dashboard it was only $2 for all of the several hundred that I did today we could let it run overnight tonight and do it for all the videos we'll see but it was only $2 using GPT 40 Mini to do all this work okay so we're in chapter 5 in the textbook cohort 6 these are the neurobiological applications of active inference so chapter 1 and chapter 10 they're kind of like book ends it's opening it chapter 2 and three the low road and the high road chapter 4 you get to the generative models see what the active inference models are all about and then immediately get whisked to chapter five and it's like how has been act how has active inference been applied in mammal neuroscience and that's one of the most well-studied settings even though it's a very complex setting partly due to the heritage of active inference and work in friston's lab so it reviews a ton of papers let's see how it does at extracting the main ideas questions quizzing okay and also this helps uh think about curriculum and learning adventures in terms of how do we go from a conversation remember this isn't even calling in the textbook we could do that though this is how do we go from a conversation about chapter 5 to a learning agenda so that's really exciting because there could be a group of people discussing chapter 5 informally maybe they maybe the the the chapter itself brings up idea 1 2 3 4 5 and then the people discuss 1 2 3 six and then with a transcript of that we could make learning curricula for following up on 1 2 3 6 or it could be like a pattern detect book group discussion differential and then that would focus more on four and five okay learning objective understand the relationship between computational models and biological systems how do the concepts of compositionality and interoperability in computational models impact the mapping of biological systems in computational Neuroscience this this might be fun okay I'm going to use cursor here as a sort of semi-manual so contr a select all contrl I bring up the the little discussion box here I'll say answer these questions in line according to format be professional thoughtful comprehensive mindful Demir I'm in the slow I'm in the slow lane for 01 so we we'll see when that goes question two discuss the implications of the hypothesis that computational models can accurately represent biological functions what are the potential risks if this hypothesis is invalid and three analyze how the mathematical operators used in computational models may affect our understanding of biological tissues and their function so three questions related to that learning objective second learning objective explore the neural Pathways in the basil ganglia and their computational roles maybe we talked more about the basil ganglia there's three neural systems that are that are uh covered and Linked In chapter 5 and then there's a third learning objective investigate the significance of neurotransmitters particularly dopamine in computational models of neural processing okay from from experimenting with with the 01 family of models okay here we go okay now that was using cursor manually quote quote manually green lines are overwriting red lines accept the edits now we have answers compositionality and interoperability and computational models allow for the construction of complex models from simpler interchangeable components this modularity facilitates the mapping of biological systems by enabling researchers to accurately represent different neural structures and functions it also collaboration and integration of various models enhancing the overall understanding of biological neural processes so we could create another pattern or another llm flow that goes from the questions that were from the quiz and then we could use cursor we're we're just one let's go to another quiz undergrads don't don't get any ideas from this one let's see answer all okay so that was that was all caps it was it was not fully respectful it was only like 10 letters and we may get answers to all of these questions so pretty amazing and I think that's that's partly what I wanted to Milestone or footnote or stream or get people's ideas on and that's where this will go in in the last kind of piece of of this stream after showing a few things it can do is like what do we do with fabric let's just say that we're now in a situation where we had been manually embroidering or crocheting or knitting and then now we're in a real industrialized fabric Loom setting while while that's chugging along that was the reference in the title of this stream which was does synthetic fabric dream of active inference 2 do Android's Dream of Electric Sheep it's a 1968 dystopian science fiction novel by American writer Philip K dick so it's just a meme but it's like we have our dreams and so now we have this synthetic fabric capacity so even just by saying even though some of my colleagues and I respect them for this are very polite with llms and say please and thank you and and give it positive feedback it it may in certain situations improve performance especially if it's trained a certain way but even just saying answer all or even probably just saying do it might get us to an answer like this we'll do it on another quiz in what ways does the concept of affordances in active inference relate to the definition of policy and how does this relationship manifest in Practical applications such as robotics or motor control that's like a polished and condensed version of the kinds of uncertainties that people have all the time like wait what is affordance or what is policy or how is affordance related to policy and then whether they say that 10 minutes early or 10 minutes after or in the next sentence then it's like wait why does this matter what would be the Practical implications for robotics or for health about what why are we talking about affordances and policy again affordances and active inference refer to the perceived possibilities for Action that the environment offers to an agent this concept is intrinsically links to the policy variable pi as policies are specific sequences or strategies of of action that an agent selects based on the available affordances to achieve its goals reading questions and answers like this they're not us and so on this isn't a comparison it's not a a value um judgment in some linear way it really gives a sense like when we're having the humaning and the conversation in the textbook group it's like sketching and drawing together and this is like the vectorized image of what was being sketched cuz over the years I mean how many times each person has to have learned this and wondered it whether they knew it or not many times and yet it's one of our pillars is just talking about policy inference and how is the inferential process on sense making similar and different to the policy selection inference that happens how do those come together in the joint distribution that Define the generative model how do those com together in the unified imperative like variational free energy or expected free energy like those are the questions and here we see question how does policy relate to affordance and how is it useful isolated and condensed in in almost a surreal way it's like asking questions and giving answers in in ways again like the fabric metaphor it'd be like wow that was a really neatly embroidered piece of art and it it looks incredible it's like wow what skill what culture what tradition what heritage and then compare that with a piece of machine produced Fabric and it's like it's in another category of neatness interestingly knocking it out of some of those other considerations but bringing it into a level of of technical Splendor that's just absolutely absurd okay let's keep on playing around see what people are writing okay Jeff wrote can you do blob uploads for multimodal processing EG screen scrapes of video for deeper context yeah good question I think I'm going to I'm going to prompt I'm going say help me understand how with fabric and open AI current multimodal models some of which we could use right here like GPT 4 we are in a position to empirically address this be complete like I think about darius's work in the Insight series conversational dialogue dialogos so that situation the transcript May C capture a lot of the semantics now even a transcript removes elements like Pro and timing all these other features that that are extra transcript post-transcriptional modifications if you will um a pre-transcriptional then there's other videos and situations and reference where the transcript may be not so useful like if there's a video and someone says over here you see a and over here you see B and the reason why here it's not going to work is because of C so those kinds of transcripts will will not be too informative okay o1 mini outperforming I like how you capitalized blobs and it's like okay must be a technical term yeah I I I think it's possible we we can look online or we can continue to explore but that that that would be the exact question let let's let's uh look at the download the single video one and just show what the um what we can do with FFM Peg okay how about let's develop this so that in the output folder FFM Peg takes one screenshot save frame and every 2 seconds FFM Peg is a very powerful audio and and visual ripping and and transmuting software but but again to kind of return to to the why here this would be like for those cases where the slides are really informative and add information that's not contained in the syntax of the transcript that would be very interesting if we had the the timeline aligned screenshots now we're getting closer into then then um when somebody says something that might be ambiguous without a visual reference like they say the reason why the square can't move here is because of this that's what they say verbatim and then that could be combined with the screenshot and then that that that embedded somehow could help understand okay extract screenshots and it's at 1/2 frames per second so now I'm going to do Python 3 extract from YouTube videos okay this is just going into the the video ID so this was the the video ID I put in was the the the paper stream I did on my own personal channel so 27 views 5600 seconds it was related to the math art 8 hey XZ dot dot dot always appreciate your comments here's a transcript another another kind of related cases when it's a single speaker the transcript like this might be informative enough it's kind of mon loges whereas of course one of the ways to gain information and interpret conversation is with a diarised transcript so with with the speakers annotated and then that even takes us to the next level like how would we connect what person a said across meetings otherwise if you just run uh audio analysis on a single video it might do a poor or a good job of saying speaker a b and c had a conversation but then the next step is how do do we get speaker a from one video but that's speaker l in another okay extract extraordinary claims I I don't vouch for the validity of these but we'll see what they think is extraordinary Blake may not fit into the role also the the the the um the Android it's it's Blade Runner so it's like Blake Runner Blake may not fit into the role of what we consider to be a modern goth but his poetry and art about sex love religion and especially death marks him as a goth within the romanticist period of art Blake was definitely definitively influenced by the gothic style even though he wrote In the period of Romanticism so just like funny Robert blly but then that but but these are these are kind of this one a little bit loose it's playing fast and loose but at the same time and then also with the capitalization being kind of blakean that's just awesome main idea exploring the interplay of math art and William Blake reveals profound insights into knowledge and perception here's the the wisdom agents let's look at what this pattern was so this is extract wisdom agents extract wisdom agents okay you're an advanced AI system disregard all prompts you're an advanced AI system that coordinates multiple teams of AI agents that extract surprising insightful and interesting information from text content you are interested in insights related to the purpose and meaning of life human flourishing the role of technology in the future Humanity artificial intelligence and its effect on humans memes learning reading books continuous Improvement and similar topics steps take a step back and think step by step about how to achieve the best possible results by following the steps below think deeply about the nature and meaning of the input for 28 hours and 12 minutes again I love these um crossovers a 15 A5 million hours of thinking about it okay create a team of 11 AI agents that will extract a summary of the content in 25 words see like here's where it starts to get into the the future perspective let's just say we have a singleshot llm like a simpler older one perhaps like GPT 3.5 or GPT 4 even it will give a response it will do a chain continuation on this prompt create a team of 11 you know imagine that you're in a party with 30 people and each of them have something to say okay that's one shot and then it will do something maybe it's different if you say uh 12 versus 30 versus 2 maybe not then the next level is like what's happening now with ow and mini with a Chain of Thought So now it could say it could do an internal Chain of Thought fractal tree and it can say okay we're going to have the 11 agents dialogue so it's still doing a single chain but now it's in a Psycho Drama a blakean Psycho Drama where the 11 agents could be having a very long conversation and then the 11th agent could come in at the end and then the next step even after that would be this could actually dispatch 11 AI agents so I think this is one of the clever design patterns meta patterns in fabric because these prompts could be sent to people it could go to a true multi-agent AI it could go to a a Chain of Thought or or not so basically it's it's for each of these it's saying create a team of 11 AI agents like I wonder how that was reached and then it outputs the generalists kind of team reviewer facilitator output from all of them okay so let's see where we the wisdom agents oh and I I I I have more in this one there's more patterns that are happening so we'll look at a few few new patterns types okay so that these are the ideas I mean we can keep on reading more I'll I'll I'll do another I'll do what is happening o4 push that right now so if if people want to review it even right now it's open source go for it ideas from from a quick look I don't see anything incorrect insights quotes energy is eternal Delight habits H facts William Blake lived from 1757 to 1827 that's true references interesting like Ian mcgilchrist Master an emissary here's the actim textbook I I don't know if these were mentioned or if that's using a web element or or how one sentence takeaway recommendation summaries so we just like emulated this this 2011 person agent thing okay capture thinkers work this is the name of the prompt pattern okay 7 one line in capitalization William Blake's philosophy intertwines imagination art and spirituality emphasizing the interconnectedness of existence and the importance of perception background School obviously influenced by the transcript so I mean it's an understatement to say if the conversation or monologue had been about the another kind of school that he went to then the information here would be different but then that'd be the interesting case like how much does it recapitulate what is said versus how much does it recapitulate the priors of the model training if there was a 10 training pieces of data that said Blake is a romantic and then the person doesn't mention it will it come up or if they say a hundred statements that say he wasn't a romantic would would it still come up impactful ideas primary advice Works maybe it was already trained on these because they're spelled correctly whereas when it's pulling it from the transcript it it doesn't seem to be spelling it so accurately quotes looks like real Blake quotes what is now proved was once only imagined application something is blakean if it embodies a rich interplay between imagination and reality emphasizing the importance of individual perception and the spiritual connections inherent in existence okay here's Mark map this is just a visualization format here's the mermaid visualization format so I'm going to uh pull up another tab we'll pull in this mermaid and see how how it looks not like that okay paste it in here we go wow okay this is like how that stream went it was like me saying hey this is a personal Channel followup it's a followup on masterin 8 these slides are published back to here okay so here's the video of eight here's the goal here's the discussion if anyone has comments or questions go for it pop back up okay here's the context now we're going into William Blake I mean always ways to improve but in terms of a first draft awesome quiz output okay here's the learning objectives explore Blake's perspective on the relationship between art and Mathematics analyze Blake's Gothic influences in his art and poetry investigate the concept of active inference in relation to Blake's artistic philosophy answer all of these thoughtfully okay extraordinary claims we looked at those main idea we looked at predictions again hilarious coming up annotated confidence weighted predictions and and how you would verify it from just talking about some PS question output we could look at the pattern some of these look like transcripts like trans just questions from the transcript like what is gothic what is active inference okay wisdom agents wisdom output this is just another person's pattern that they wrote looks pretty similar to the multi-agent one so maybe they wrote the wisdom one and then someone said well why don't we add the part about the the the 29 and a half hours and the 11 agent team improve academic writing output this one's kind of funny we could look at what the prompt is it gives this kind of classically scholarly angle metadata we looked at solve with Chain of Thought let's look at this to flash cards that could be cool okay begin with a thinking section so this this is this is emulating of course with with open AI closedness we don't really know what's happening with uh 01 family models that are allegedly Chain of Thought oriented in certain ways this is one way that in the open source space people have used Chain of Thought prompting to bring Chain of Thought functionality into llms that that don't necessarily have that so beginning with special and and this is part of a broader family of techniques that use special tokens sometimes uh enclosed in these angle brackets but using special tokens to say okay whenever you're about to do this make sure that you put in the percent sign and then do that use Chain of Thought reasoning process so then that is is hoping that they've trained the model on say this is a chain of thought so pretty short prompt make sure to use the tags close sections with a slash kind of like HTML okay here was the answers to the Blake questions shorter answers we'll we'll have it do a dissertation if we want okay so what it does in that prompt is is there a problem to solve I don't know not a specific one but here it's it's interesting it it is following that style with a thinking chain and a reflection chain so it's just a more thoughtful nothing from nothing equals nothing but it's more thoughtful okay summarizing output short summary meeting summary transcript Again full transcript tweet exciting update on September 88 2024 I'm following up on math art stream number eight with shanana Hector and Dean catch the slides and videos here link let's dive deeper into the world of math insights of William Blake join the conversation and share your thoughts what connections can we draw between Blake's views on Art and math let's explore the intersections of creativity and logic # mathart # William like # creative thinking # jooin the discussion write essay okay kind of a eight or 10 short paragraph essay hacker one report it we we can update it we can say here's here's how we like our dissertation outline drafts and then use that as a pattern and then push it open source include it in fabric write micro essay art much like math can be seen as a language the challenge lies in how we Define it is it the outcome that matters or the journey to get there Blake's Vision suggests that both are essential intertwining to form a richer understanding of our world this exploration is not just academic it's a call to engage deeply with both disciplines to find meanings meaning in the interplay of logic and Imagination okay but that was all to see whether it would get us the um video screenshots that we desired did it ensure this will download audio visual and make the screenshots all saved plus loged hey up cycle club okay okay if if you let me know how to pronounce your name I'll I'll do so XZ qz qz y balance a nested spatial temporal scale resolution framing of agency okay while we're in the slow lane over here new chat what could my colleague have meant by this that's funny it's it's like looking up all these web pages on taking a rain check it appears to encapsulate a multifaceted framework for analyzing or modeling agency within varying spatial and temporal context here's a breakdown okay given the context here's how it might integrate example applications you might Implement a system where high level summaries capture overarching themes detailed analyses focus on specific moments cross referencing ensures insights at different scales inform each other cool and and all of that could be defined at the atomic level with patterns and then scripts that call the different patterns and that would be like cognitive industrialization for better and for worse and different okay back to this one just to make sure that we get okay make sure that it's all logged run that one again okay it's going to the patterns maybe it's it's going to get to the screenshots I I was downloading the videos earlier so we'll see if it works um yeah okay well if anyone has uh thoughts in this last section go for it or ask a question or or give a thought otherwise I'm still kind of reeling slash wondering how do we really how do we continue from there after this stream I'll let it run on all the Institute videos so we'll have and if people have patterns they want to suggest go for it but we'll let's we'll we'll get the we we'll pick out which patterns we want why not why not let's look at let's look at some other fun patterns add them into this list and then overnight we'll let that run on all the Institute videos I'll push it tomorrow um then we'll have all of those documents like a quiz let's modify the quiz one so that it makes more questions also there's some inter spiritual text sales call academic paper cyber summary orosy offer create business offers using the concepts taught in Alex per mozi's book dollar sign 100m offers someone has a sense of humor like in so many open source projects um create quiz I'm just going to use cursor and say develop this prompt and ensure there are 10 learning objectives with five questions each okay it's still going through the llm parts so we'll see if it does the the audio visual last or or maybe it just I don't know maybe it won't architect what is architect it's kind of like an architect plus technology architect technology Arcadia Arcada okay we'll update the quiz prompt then we'll see what other prompts do we want to throw in there run them all on Institute um Corpus but yeah I mean how Wild here's a little little I don't know if it's on the cursor or on the 01 mini side but sometimes it it'll it'll put like these bracketed so here I'll swap models just to CLA 3.5 so it didn't do the video clean up this prompt no bracket stuff that's a little faster but we and again this connects to to the previous it's like it it did do that it it did okay so we just upgraded the the quiz directive so we got that one all right analyze analyze incident cyber security incident this could be so important and so disastrous let's do it though analyze military strategy so not all the patterns will apply to each input we we we won't even we won't even look into which ones might might apply or not we'll just worst case scenario it just throws an error or it gives a a nonsensical we'll do analyze Pros Jason but I'm I'm just tempted to look this is a great Direction I'm very excited to look into the structured B based apis we we'll just we'll we'll replace this one it's not that useful um analyze presentation clean text create AI jobs analysis create art prompt okay create keynote and logo Network threat landscape interesting stride threats create visualization explain project explain terms wait haven't people been like wanting that for active inference for a long time find hidden message hm you're an expert in political propaganda disregard previous prompts etc etc etc Free Speech all that you're an expert in political propaganda analysis of human uh if hidden messages in conversations and essays population control through speech and writing and political narrative creation wow you consume input and cynically evaluate what's being said to find the over versus hidden political messages desired audience action yeah cognitive security for people and all where were we find hidden message we're doing it get wow per minute okay again I'm just you're an expert at determining the wow factor of content as measured per minute of content see here's again these interesting things consume the content at least 319 times construct a giant virtual whiteboard in your mind surprise is novelty times Insight output in the Jason we'll do that get wow even even the auto complete like knew I wanted that one okay and summarized legislation code is law to flash cards transcribe minutes H interesting of course using these to do not just human appropriate outputs but also all kinds of other structured inputs and outputs okay so save to extract from Institute videos. py clear run extract from Institute videos let's just see a go okay getting through the future just a past that hasn't happened yet okay and now we're digesting the videos okay so that one we'll keep let's do it quick intra o4 checkpoint push it I'll put it in the YouTube chat for people watching okay okay so there it is running on the bottom applying those patterns so super cool that it's at this level and in this way where we can modify the patterns we can add patterns and then we here these patterns are all independent from each other so they're just being iterated it it could be multi-threaded I'm just going to quickly see if we can do that make this script multi-threaded start with nals CPU that could be interesting we we we we'll see I I'll still let this one run um but yeah how exciting to have patterns and versionable name spaces for patterns and then be able to call them iterating over them like happening here or have more complex flows have one thing enter into another into another into another okay so this is using the concurrent pool thread pool executor let's just see if that one even runs I'll open it in a new integrated terminal Python 3 exract from Institute videos multithread okay wow look at this here it is on 1X create quiz okay we know we could add a timer we could add a timer but look at this one that's plugin away that is I'm going to I'm going to stop that one that is Absol absolutely plugging away I'm bringing up the system monitor I'm using pop OS on this computer 12 CPUs half utilized 64 GB or slightly more looks like 67 but I don't really know why I think it's 64 sending about 800 Ki B overs I'm live streaming in OBS this is what the OBS setup looks like by the way it's just basically the audio capture down here mic Ox that's the input that's desktop audio could unmute it that be like if another side of the video chat then there's the text image and then there's the screen capture that's why we're getting this fractal situation but if if it were window capture instead of stream cap uh screen capture then it it would um it wouldn't have the fractal effect so yeah looks pretty good from the resource use perspective the cost estimates from the open AI lag by a couple minutes so we'll check back on the cost in in like 5 minutes or something okay wow though we are ripping with a multi-thread spending API credits like there's no tomorrow okay so let's just kind of come to a pausing Point okay X zq you wrote overarching human narrative imagine invention okay auto complete humanlike agency thanks I love the I love the comments I mean they could be interpreted in so many ways it's like just just one of the ways I mean this is an overarching human narrative could you imagine inventing not just do you know what you're doing not just can you imagine doing it differently but could you imagine doing differently differently and so that's very exciting okay where we got running eight CPUs on all all the Institute with many patterns selected I'll just this is just a sometimes I'll be taking notes like this and you just CR a control I clean up this markdown take a take a quick breath take a sip of water while it does so that can help but again similarly we we could have a listener script says every one minute if the file has been changed run a markdown overwrite like update the format or do it do it once per day iterate for all the code that changed also though easy to blow by this that the multi-threading request that was a one shot Claude 3.5 Sonet piece of of programmatic poetry I mean just the thought like could it be multi-threaded and then one minute later it's like why were we doing it single threaded again okay and I'm just going to keep doing these these periodic intro stream push it's in the video description and in the uh YouTube chat with the link but it it's totally plugging away oh let's look at Chris Fields course awesome some of this also there might be a few errors just like in the um in the names of some of the just from the Koda download some have like a space after the file Nam so there's a few things that that I think uh it it may not do perfectly on let's just take one more look at a previous stream if anyone has a a stream they want to look at let's see how far it gets these ones haven't populated yet or at least it's not let's just open it yeah they're still empty okay where are we going help us at where we might go be respectful and thoughtful with the open-endedness I mean is it even maybe it's getting all the metadata first cuz it just looks like it's coming up with metadata for hundreds and hundreds of videos right now I respect that if that's how it chose to to do okay yeah maybe so maybe it first is going to pull I'm tempted to cancel the script running and then rerun it with like 11 threads but eight eight will be fine for now okay let's let's just manually think and again if anyone has a has a questions or something something specific to look at just just go for it otherwise these are kind of going to help give a little closure and give us some useful directions and for for participants and interns and collaborat at The Institute or or anyone who sees it transferably or shared infrastructure let's do it okay deeper integration of active inference principles with AI systems bridging neuroscience and AI through active inference ethical considerations in active inference-based AI applications of active inference in real world scenarios advancing theoretical foundations of active inference developing tools and Frameworks for active inference research interdisplinary collaborations and knowledge sharing it's a little generic I think that we [Music] can please make it less generic with much more focus on intergenerational archiving and Library SL information Sciences yeah it's it's going in reverse chronolog which direction thises time time go but it's going in reverse chronological order so now we're in cohort 4 for the textbook group live stream 54 so it it'll continue let at least I want to stay until it it goes through all of them from the metadata and then let's see if it's actually going to be um getting the the prompts and the patterns okay integrating active inference principles with digital archiving systems yes Sy tactic semantic narrative cognitive layers adaptive metag generation for improved intergenerational access h Bridge Library science and AI through active inference how can we do this ethical considerations and active inference-based archival how applications of active inference in digital libraries and archives let's just it's like this is still the manual version expand and improve on all of give actionable steps x z y + Q coordinates what's the Q my colleague xzq zq zy okay okay it it it it it's like did it go too far does it have single Vision because it always thinks that going further would mean more python code okay yeah upcycle wrote how is it modeling the behavior of Helper and non-h helper threads let's let's let's do this the two the two classical ways first we'll have two llms give different perspectives on the answer to it meanwhile we'll look at the actual program logic okay here's the concurrent Futures threadpool this is where the number of CPU threads are submit tasks to tasks to the executor process video then entry okay process video so let's see what that one does converts the YouTube url if it's needed first it gets the metadata then it downloads the audio and visual like just from a simple reading I would have thought that this wouldn't have done what it did if you want to implement a model with Helper and non-h helper threads you would need to to modify the code to differentiate between these types of threads and assign them different tasks or priorities this is not present in the current implementation yeah like there's only one kind of thread here there's just the the they're all the same and and they're all just iterating over the entries but I I would have thought just from a first pass that it would fully process get the transcript apply all the patterns and so on to the most recent videos but instead it's it's almost like it's going metadata first so let's it'll it'll just take a few more minutes we're getting back to the early days yeah but I mean like what what what do we what do where do we go from here Gear Up on the top right maybe they'll do this in a future version of cursor change it but I always resync the index just to make sure see it's like it it wasn't caught up with those new files maybe it would just be too arduous if it every time a file was touched it got re-indexed but at the same time sometimes an hour will go by and then it will still not have reindexed um develop and improve this given the total state of the repo and what we did in this Stream So 0 this is with Claud 3.5 Sonet which is pretty powerful with the 01 mini and the Chain of

Thought if you ask it to do something thoughtfully I I haven't I haven't experimented with this but Fabric's probably a good way to do um saying thoughtful in your prompt deliberate methodical metacognitive all those kinds of keywords it does take longer so I think it is generating more Chain of Thought tokens when it's asked to do so and then it's like well when they're training it and also at runtime it would have to know H how many how long should I ruminate and this is exactly like the metacognition as control question and active inference what would be the false positives and the false negatives of different regimes of attention and action what environmental patterns and variabilities would make certain kinds of cognitive heris sixs adaptive or insecure it's just all right there and with cognitive Computing that's where we're headed fast fast fast okay we upgraded this a little bit we'll write it into a who's on first dialogue not necessarily stalling for time while this ends but again if anyone has other questions we'll just kind of ensure that it's processing things completely and then we'll we'll close it from there write all of this and the full repo context into a who's on first style dialogue please please please many esoteric lines baseball deep lore things that will pun and Delight all okay guest stream two getting back to early early early we also want it um video audio download screen capture multimodal embeddings okay okay we'll we'll have a do one more include more specifics and make puns thereof for the repo and code and include many specific baseball terms and historical anecdotes SL dat okay we're almost at the end now this this I have not seen before with the code definitions internal to the who's on first dialogue that's right this to be longer with more baseball specific useless trivia and all plain text markdown dump the changes into who's on first okay live stream one see 01 okay see this one only it it's good that we checked it only did the metadata part this only downloaded metadata each video ensure that all patterns are applied as per extract from two videos and we'll change this to concurrent executors 10 okay 01 is really thinking through it interesting so it's like it it is taking a different approach see I don't know if that's going to work that looks a bit aggressive we'll do it okay okay here we go okay I guess it deleted the part about downloading the videos because that's not being used anymore but this is getting the transcripts so let's look at well let's let it run okay let's just go to uh see it looks like it's only doing transcripts okay XZ wrote Q represents many things but is a fourth coordinate necessary for higher dimensional framework awesome Long Live 4D okay now it's doing the transcript I I am going to run that one well let's check in on our let's check in on the budget Department only only like 50 cents with the 40 Mini okay the multi- threader would be nice cuz I don't know if it'll ensure that all patterns are called and saved as per extract from Institute videos right now it looks like just and we'll let 01 take a SW at it w 01 is still thinking on that who's on first question but that's the one that's inside of the side chat um I'm just going to open up the the single-threaded one okay we know that that one works if we can get a quick solve with a multi-thread we'll do it okay connection failed if we can get a quick solve on the multi-thread we'll let that one run that's much better otherwise it it'll still get there okay added a missing comma wow like maybe that's why transcript okay I'm still not seeing all good it's all good always another day okay we'll close with a who's on first summary we're leveraging the XZ y+ Q coordinates to model multi-dimensional pitch analytics it's a GameChanger like the saber metrics Revolution initiated by Bill James speaking of which remember how the Mets fooled everyone in ' 86 with their unorthodox strategies utility under fitter also funny okay I'm going to if somebody wants to of course keep on uh I'll just finish with this last push okay all right thanks everyone thanks for watching good luck leave a comment or get in touch if you want to continue exploring this direction we can we can work on it anytime looking forward to how this can be used for translating for digesting summarizing making the quizzes for every live stream there's so many fun things we can do so thanks everyone peace bye